

Unit 2, Lesson 1: Applying Product Laws of Exponents to Algebraic Expressions



Benchmark

MA.912.NSO.1.2 Generate equivalent algebraic expressions using the properties of exponents.



Learning Target

- ☐ I can generate equivalent algebraic expressions using the properties (laws) of exponents to find the product of monomial expressions.



2.1.1 Warm-Up: Growth Mindset

- Write a check mark ✓ in the box that best describes how you feel about mistakes.
 - ☐ I do not like to admit I made a mistake.
 - ☐ I think all mistakes are bad.
 - ☐ Mistakes tell me that I won't ever understand.
 - ☐ Mistakes make me wonder what I am missing.
 - ☐ I think a mistake is a chance to try another strategy.
 - ☐ Mistakes help me learn.

- Consider the acronym shown.

How does this interpretation compare to your feelings about making mistakes?

Means

I

Start

To

Acquire

Knowledge

Experience

Skills



2.1.2 Guided Instruction: Equivalent Algebraic Expressions

The laws of exponents can be applied to generate equivalent algebraic expressions.

- The table shows three expressions and two different methods that can be used to determine an equivalent expression.

Expression	Expanded Form	Product of Powers Law	Equivalent Expression
$x^2 \cdot x^3$	$x \cdot x \cdot x \cdot x \cdot x$	x^{2+3}	x^5
$-4x^2 \cdot 5x^3$	$-4 \cdot x \cdot x \cdot 5 \cdot x \cdot x \cdot x$	$-4 \cdot 5 \cdot x^{2+3}$	$-20x^5$
$6x^2y^0 \cdot -xy$	$6 \cdot x \cdot x \cdot 1 \cdot -1 \cdot x \cdot y$	$6 \cdot -1 \cdot x^{2+1} \cdot y^1$	$-6x^3y$

- Discuss with your partner any patterns you notice between the expanded form and applying the product of powers law.

- Complete the statements.

When multiplying monomial terms, the coefficients

of the terms can be ☐ added ☐ multiplied to find the

coefficient of the equivalent expression. The

product of powers law of exponents can be

applied to ☐ bases ☐ coefficients that are the same.

- Complete the multiplication of $6b^7 \cdot 8b^{11}$ to find an equivalent monomial expression.

$$(6b^7)(8b^{11}) = (6 \cdot 8)(b^7 \cdot b^{11}) = \underline{\hspace{2cm}}b^{\underline{\hspace{2cm}}}$$



Product of Powers

$a^m \cdot a^n = a^{m+n}$ and conversely $a^{m+n} = a^m \cdot a^n$

- Determine an equivalent monomial algebraic expression by filling in the blanks.

$$(3a^3b^2)(7a^{12}b) = (3 \cdot 7)(\underline{\hspace{2cm}} \cdot \underline{\hspace{2cm}})(\underline{\hspace{2cm}} \cdot \underline{\hspace{2cm}}) = \underline{\hspace{4cm}}$$

- Abigail and Mia both attempted to write an expression equivalent to $30x^7$.

Abigail	Mia
$30x^7 = (6x^2)(5x^5)$	$30x^7 = (3x^4)(10x^3)$

Discuss with you partner whether or not both students generated equivalent expressions. Write a summary of your discussion.



2.1.3 Guided Practice: Products of Monomials

1. Generate an equivalent algebraic monomial expression.

a. $(-9t^0u^4)\left(\frac{1}{3}u^{11}\right)(-10t^{20}u^8)$

b. $(9q^{-3})(0.6q^{10})(12q^5)$

c. $(-4x^0y^2)(-10x^3y^5)(1.2x^4)$

d. $(1.2m^2n^3)(-2.25m^7)(-0.5n^{12})$



2.1.4 Collaboration: Raising a Monomial to a Power

Additional laws of exponents can be applied to generate equivalent algebraic expressions.



Power of a Product

$$(ab)^m = a^m \cdot b^m \text{ and conversely } a^m \cdot b^m = (ab)^m$$



Power of a Power

$$(a^m)^n = a^{m \cdot n} \text{ and conversely } a^{m \cdot n} = (a^m)^n$$

1. Two students began to generate an equivalent expression to $(-2x^4y)^3$.
 - a. The beginning of each student's work is shown. Finish each student's work.

Kaniya's Work	$(-2x^4y)^3$ $(-2)^3(x^4)^3(y)^3$ $= \underline{\hspace{2cm}}$
Nicholas' Work	$(-2x^4y)^3$ $(-2x^4y)(-2x^4y)(-2x^4y)$ $(-2 \cdot -2 \cdot -2)(x^4 \cdot x^4 \cdot x^4)(y \cdot y \cdot y)$ $= \underline{\hspace{2cm}}$

- b. Discuss with your partner how Kaniya's and Nicholas' approaches to generating an equivalent expression were different but resulted in the same answer. Write a summary of your discussion.

2. Use the expression $(12r^5)^2 \cdot \left(\frac{1}{4}r^7\right)$ to complete the following.

- a. Find an equivalent expression. Show your work.

- b. With your partner, compare and discuss the methods you used to create an equivalent algebraic expression.



2.1.5 Your Turn!

1. Determine an equivalent monomial expression for each expression using the laws of exponents. Show your work.

$(-3w^3)^2(20z^9)(-7z^{14})$	$(23k^6)^0\left(\frac{2}{3}k\right)(k^{14})$
$(27b^{30})\left(\frac{1}{3}b^9\right)$	$(6q^3s^7)^2(-q^2s^{-1})(-7q^{10}s^{14})$
$(-11bc^4)^3$	$(-4x^2y^{12})^{-2}$



2.1.6 Wrap-Up: Cool Down

Determine an equivalent expression for each algebraic expression using the laws of exponents. Show your work.

1. $(56g^7)\left(\frac{4}{5}g^{-2}\right)$

2. $(-x^2y)(7.3x^8y^{11})(0.1x^9y^3)^2$



Unit 2, Lesson 2: Applying Quotient Laws of Exponents to Algebraic Expressions



Benchmark

MA.912.NSO.1.2 Generate equivalent algebraic expressions using the properties of exponents.



Learning Target

- ☐ I can generate equivalent algebraic expressions to find the quotient of two monomials using the properties (laws) of exponents.